

• OPEN TRACK

GRAILWATCH

GrailWatch

Proof-of-reserves for zkLTC.

LIVE ON LITVM LITEFORGE · CHAIN 4441

grailwatch-mauve.vercel.app

01 The Problem

LitVM's entire thesis rests on one invariant — **1 zkLTC = 1 LTC**

Today it's "trust us"

That 1:1 backing is asserted, not proven. There's no live, public, independently verifiable check anyone can run right now.

PoR is table stakes

After the 2022 collapses, proof-of-reserves became mandatory for exchanges. The bridge LitVM is built on needs exactly the same accountability.

If the peg ever breaks, everyone should find out instantly — not eventually.

02 The Solution

A **quorum of independent attestors** observes both chains and posts the backing ratio on-chain.

- **Median, not trust.** Health is the *median* of fresh readings, so no single attestor — or liar — can fake "FULLY BACKED".
- **Instant alerts.** A subscribable `BackingAlert` fires the moment backing drops below 1:1.
- **Real, verifiable numbers.** Every figure traces back to a public source anyone can re-check.
- **Public good.** Zero-wallet status page that every LitVM dApp benefits from.

03 How It Works

1 · read

Observe both chains

Real LTC reserve (public Litecoin address via litecoinspace.org) + real live zkLTC supply (LiteForge Blockscout).



2

attest()

Each attester posts its reading on-chain.



3

health()

Returns the MEDIAN across distinct, fresh attestors — must be $\geq$ quorum and newer than maxAge.



4

events

StatusChanged / BackingAlert emitted.

consume

Public dashboard + subscribe

Anyone reads the live backing badge or subscribes to alerts. WebSocket pushes updates in real time.

fresh

Serverless cron

Keeps the feed fresh so the median always reflects current, unstale readings.

04 Live & On-Chain

LIVE DASHBOARD

grailwatch-mauve.vercel.app

- REAL DATA – NO MOCKS

qa-grailwatch.js cross-checks the on-chain numbers against the real external sources — an **exact match**, proving nothing is fabricated.

ReserveAttestationV3

VERIFIED

0xf099F039f8206C4C2BF91120A913a1F138fBAfcB

QUORUM 2

MAXAGE 172800S

MININTERVAL 30S

CHAIN 4441 · LITVM LITEFORGE

05

Under the Hood

63

PASSING TESTS

2

ADVERSARIAL AUDIT ROUNDS

V3

HARDENED CONTRACT

Evolution V1 → V3

V1 single-attestor trust → V2 quorum + median + freshness
→ V3 anti-DoS attestor cap, honest 3-state status, and a
`minInterval < maxAge` guard.

Proof, not promises

Quorum-median-freshness on-chain, real-data cross-check
in QA, WebSocket live updates, neobrutalism React UI. 63
tests across 2 audit rounds.

06 Why It's Hard Money Web3

GrailWatch directly verifies the "1 zkLTC = 1 LTC" hard-money invariant the whole ecosystem depends on.

Guards the peg

The hard-money promise of LitVM is only as good as its backing. GrailWatch makes that backing continuously, publicly provable.

Public-good infra

Not a single dApp — shared transparency infrastructure that every LitVM application can rely on and link to.

07 Judging Criteria

Innovation

First proof-of-reserves / bridge-transparency infrastructure for LitVM.

Hard Money

Verifies the core hard-money invariant the entire ecosystem rests on.

Technical

Quorum-median-freshness, 2 audit rounds, real-data cross-check, 63 tests.

UX

Zero-wallet public status page, instant backing badge, one-click independent verification.

08

Live Now + Roadmap

• Live now

- ☐ V3 contract on Chain 4441 (verified)
- ☐ Quorum-median backing health from real LTC + zkLTC data
- ☐ BackingAlert events + subscribable public dashboard
- ☐ QA cross-check proving exact match vs external sources

→ Roadmap (disclosed limitation)

The attesor set is **owner-curated** today — the median protects against a rogue attesor among honest peers, not a malicious owner.

- ☐ Permissionless attesor **staking + slashing**
- ☐ SPV proof of the Litecoin locked balance → full trustlessness

GRAILWATCH

Proof-of-reserves for zkLTC.

LIVE

grailwatch-mauve.vercel.app

GITHUB

github.com/PugarHuda/grailwatch

BUILT FOR THE LITEFORGE HACKATHON 2026